

Cyber Security Laws + Security with Kali Linux

Assignment + Viva Preparation

This document contains beginner-friendly explanations, point-wise answers, important definitions, and viva questions & answers for all assignment topics.

1. Importance of Cyber Security Laws

Cyber security laws are rules made to protect computers, networks, and user data from cyber crimes.

Important Points:

- Protects personal and financial information
- Prevents hacking and online fraud
- Punishes cyber criminals
- Increases trust in online systems

Example: Bank account hacking or phishing attacks.

Viva Questions:

Q. What are cyber security laws?

A. Laws that protect digital systems and data.

Q. What is cyber crime?

A. Illegal activity done using computers or the internet.

2. Difference Between GDPR and CCPA

GDPR and CCPA are data privacy laws used to protect user information.

GDPR:

- Used in Europe
- Strict privacy rules
- Requires user consent

CCPA:

- Used in California, USA
- Gives users control over their data
- Allows users to opt out of data selling

Viva Questions:

Q. What is GDPR?

A. European data privacy law.

Q. What is CCPA?

A. California consumer privacy law.

3. Ethical Hacking vs Malicious Hacking

Ethical hacking is legal hacking done to improve security, while malicious hacking is illegal.

Ethical Hacker:

- Works with permission
- Finds vulnerabilities
- Protects systems

Malicious Hacker:

- Hacks illegally
- Steals data
- Damages systems

Viva Questions:

Q. What is ethical hacking?

A. Legal hacking for security testing.

Q. What is a vulnerability?

A. Weakness in a system.

4. Five Phases of Penetration Testing

Penetration testing checks the security of systems.

Five Phases:

1. Reconnaissance – Collect information
2. Scanning – Find vulnerabilities
3. Gaining Access – Exploit weaknesses
4. Maintaining Access – Check persistence
5. Reporting – Create security report

Viva Questions:

Q. What is penetration testing?

A. Security testing using ethical hacking techniques.

Q. Which tool is used for scanning?

A. Nmap.

5. What is Kali Linux?

Kali Linux is a Linux operating system used for cyber security and penetration testing.

Features:

- Pre-installed security tools
- Free and open source
- Used by ethical hackers

Popular Tools:

- Nmap
- Wireshark
- Metasploit
- Burp Suite

Viva Questions:

Q. What is Kali Linux?

A. An operating system for cyber security testing.

Q. Is Kali Linux free?

A. Yes.

6. Role of Nmap in Network Security

Nmap is a network scanning tool used to discover devices, open ports, and vulnerabilities.

Uses of Nmap:

- Network discovery
- Port scanning
- Service detection
- Vulnerability detection

Practical Example:

A company scans its server to find risky open ports.

Viva Questions:

Q. What does Nmap stand for?

A. Network Mapper.

Q. What is port scanning?

A. Checking open ports on a system.

7. Metasploit and Burp Suite

Metasploit and Burp Suite are penetration testing tools.

Metasploit:

- Used for exploitation
- Finds vulnerabilities
- Delivers payloads

Burp Suite:

- Used for website testing
- Intercepts requests
- Detects SQL Injection and XSS

Viva Questions:

Q. What is Metasploit?

A. A penetration testing framework.

Q. What is Burp Suite?

A. A web application security testing tool.

8. Logging and Monitoring in Cyber Security

Logging records system activities, while monitoring checks systems for suspicious behavior.

Importance:

- Detects cyber attacks
- Helps investigate incidents
- Improves security
- Prevents data breaches

Kali Linux Tools:

- Wireshark
- Nmap
- Nikto
- Lynis

Viva Questions:

Q. What is logging?

A. Recording system activities.

Q. What is monitoring?

A. Continuously checking systems for threats.